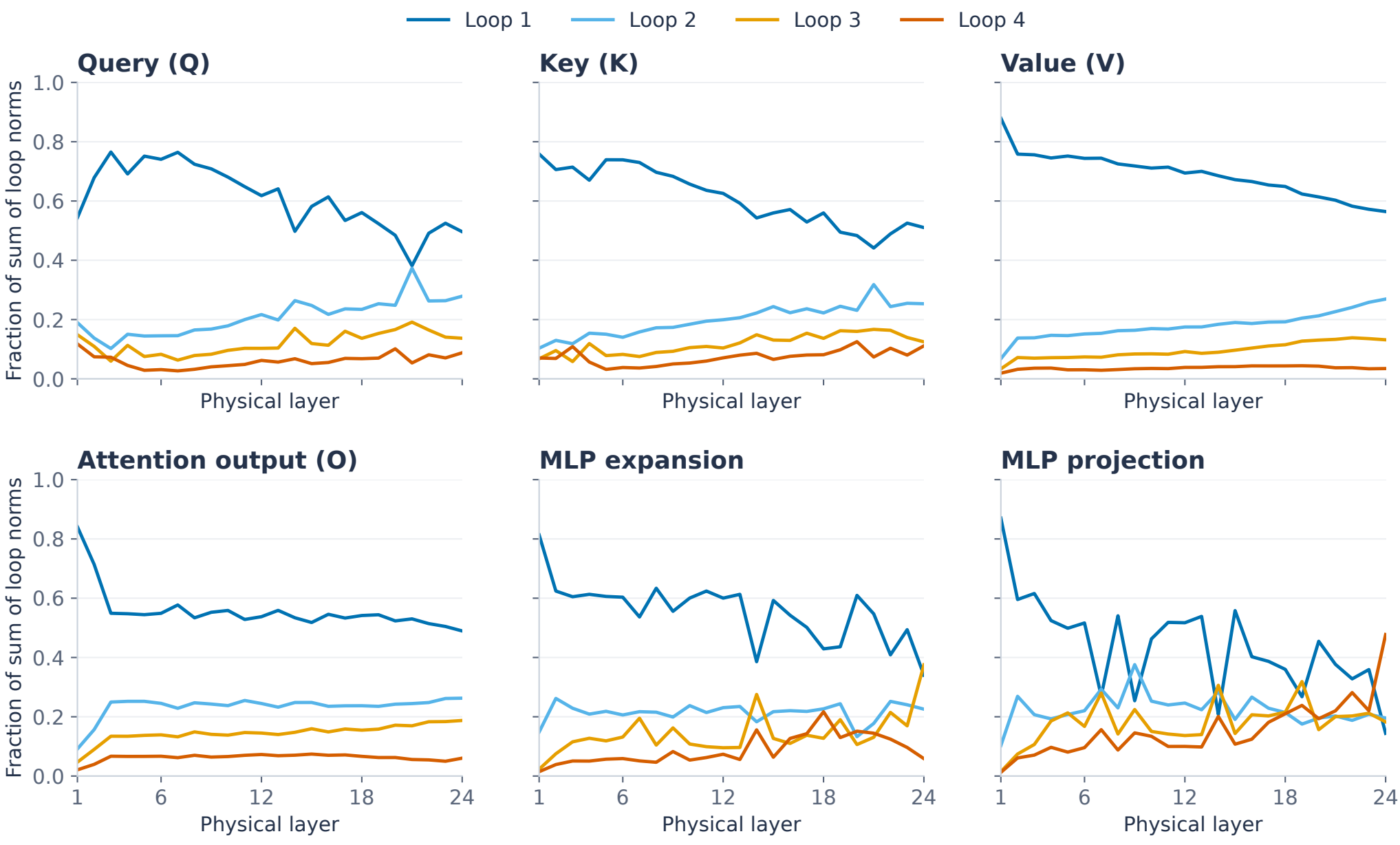
Which loop contributes the largest gradient norm?
Loop 1
Loop 2
Loop 3
Loop 4
Query (Q)
Key (K)
Value (V)
Attention output (O)
MLP expansion
MLP projection
Fraction of sum of loop norms
Physical layer
0.0
0.2
0.4
0.6
0.8
1.0
1
6
12
18
24
Average of per-batch norm shares · Four shares sum to one at each matrix
Norm magnitude is not a measure of causal importance.